Subject 65

The input files are located in the *dataIN* directory.
All the output files will be saved in the *dataOUT* directory.

Requirements

In the *NatLocMovement.csv* file, there are data on the natural movement of the population at locality level. The variables are as follows:
Siruta - the Siruta code of the locality;
City - the name of the city;
Marriages - number of marriages;
Deceased - number of deceased
DeceasedUnder1Year - number of deceased under 1 year;
Divorces - number of divorces;
StillBirths - number of stillbirths;
LiveBirths - number of live births.

The *PopulationLoc.csv* file contains the population by localities and the county codes for each locality.

1. The rate of natural increase at the county level should be saved in the *Requirement_1.csv* file. The rate of natural increase is the difference between the birth rate (number live births per 1000 inhabitants) and the death rate (number of deceased per 1000 inhabitants). For each county, the county code and the rate of natural increase will be saved. (**2 points**)

2. Calculate and save in the *Requirement_2.csv* file for each county the localities where the rates (indicator values per 1000 inhabitants) are the highest. For each county, the county code, and the names of the localities with the maximum values will be displayed. (**1 point**). Example:
County,Marriages,Deceased,DeceasedUnder1Year,Divorces,StillBirths,LiveBirths
ab,Ciuruleasa,Ohaba,Posaga,Oras Campeni,Valea Lunga,Scarisoara
…

DataSet_65.csv file contains records for 35 countries around the world regarding the population health state. The observed variables are as follows:

1. Total population – POP
2. Gross domestic product per capita (US$) – PIB
3. Health public spending (% din PIB) – CHS_PUB
4. Health private spending (% din PIB) – CHS_PRIV
5. Death rate (at 1,000 inhabitants) – RM
6. Natality rate (at 1,000 inhabitants) – RN
7. Life expectancy (years) – SPV

3. Standardise the matrix of the observed variables and saved it in the *Xstd.csv* file. Classify the observations from the *DataSet_65.csv* file using hierarchical cluster analysis: clustering method "*ward*" with "euclidean" metric. Classification is carried out on the standardised matrix of observed numerical variables and the hierarchy matrix is printed at the console. (**2 points**)

4. Determine and write in the console the threshold value and the junction at which the maximum difference is determined for the previous classification. (**2 points**)

5. Plot the dendrogram graphic for the previously determined observation classifications, along with the horizontal line at the threshold value to emphasise the maximum stability partition. The country names will be rotated 45 degrees. (**2 points**)